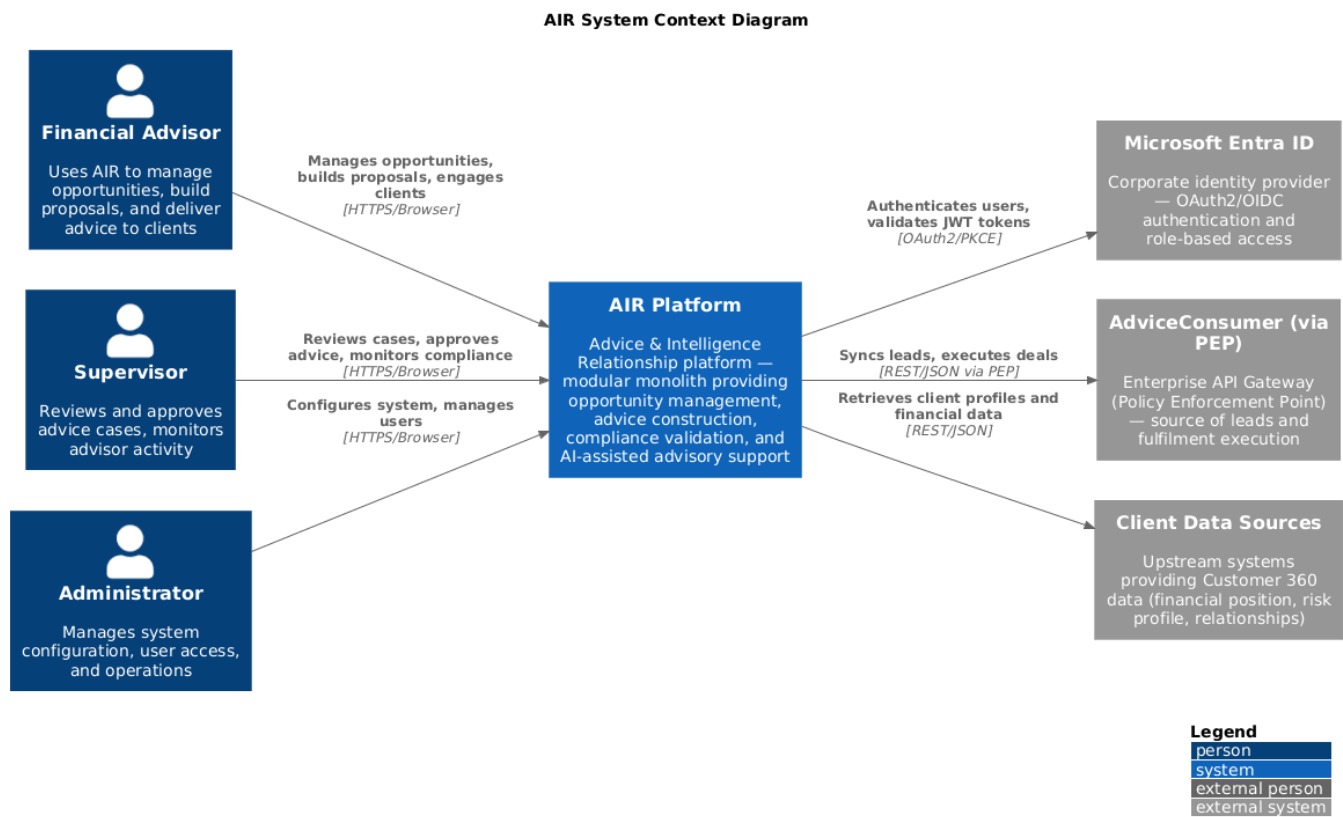


AIR (Advice & Intelligence Relationship) System Architecture Design

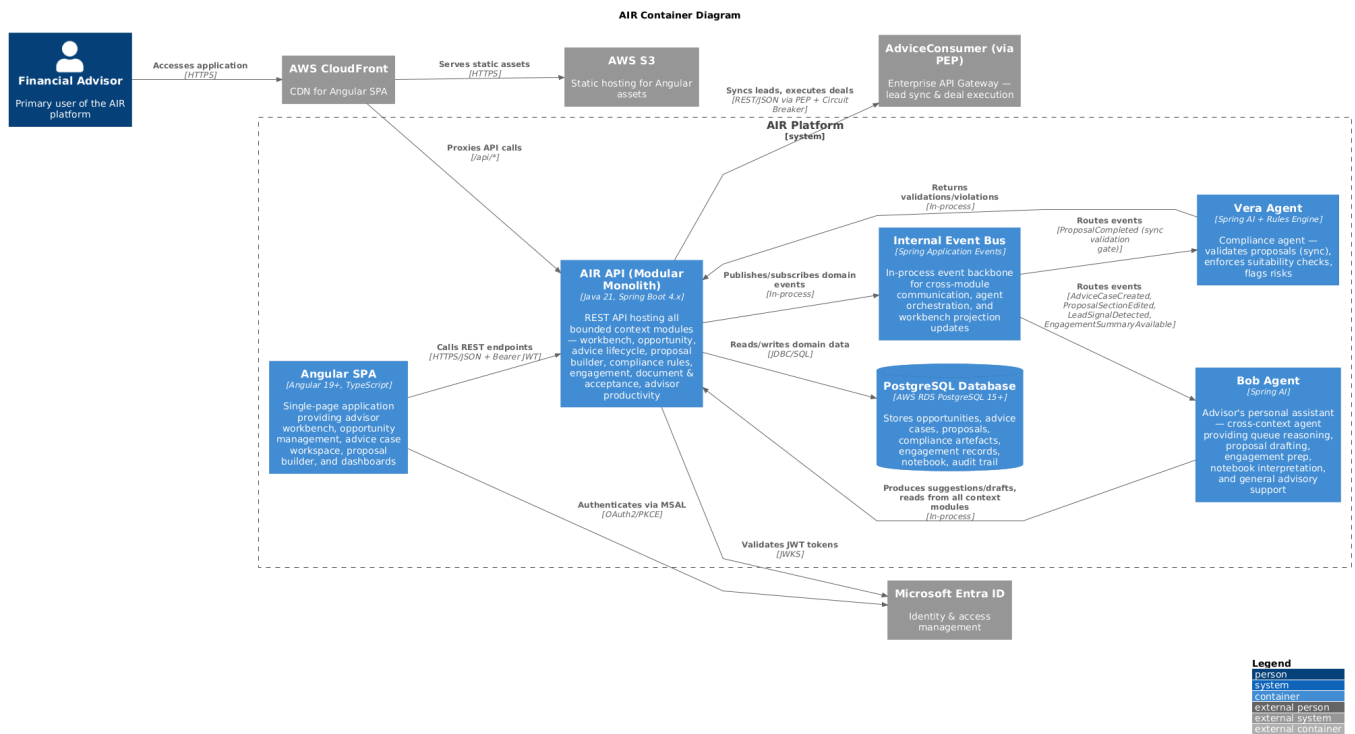
Level 1: System Context Diagram

Shows the AIR system in its environment, with external actors and systems.



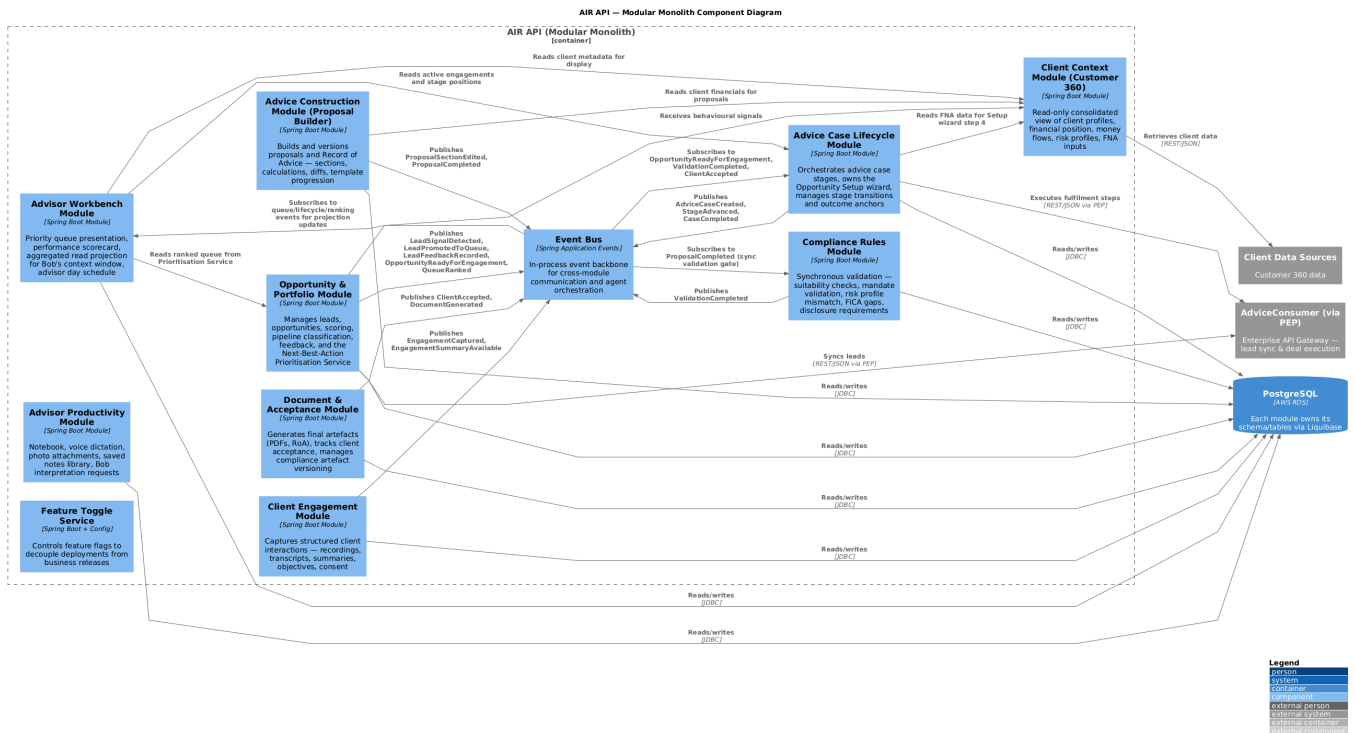
Level 2: Container Diagram

Shows the major deployable units and their interactions.



Level 3: Component Diagram – AIR API (Modular Monolith)

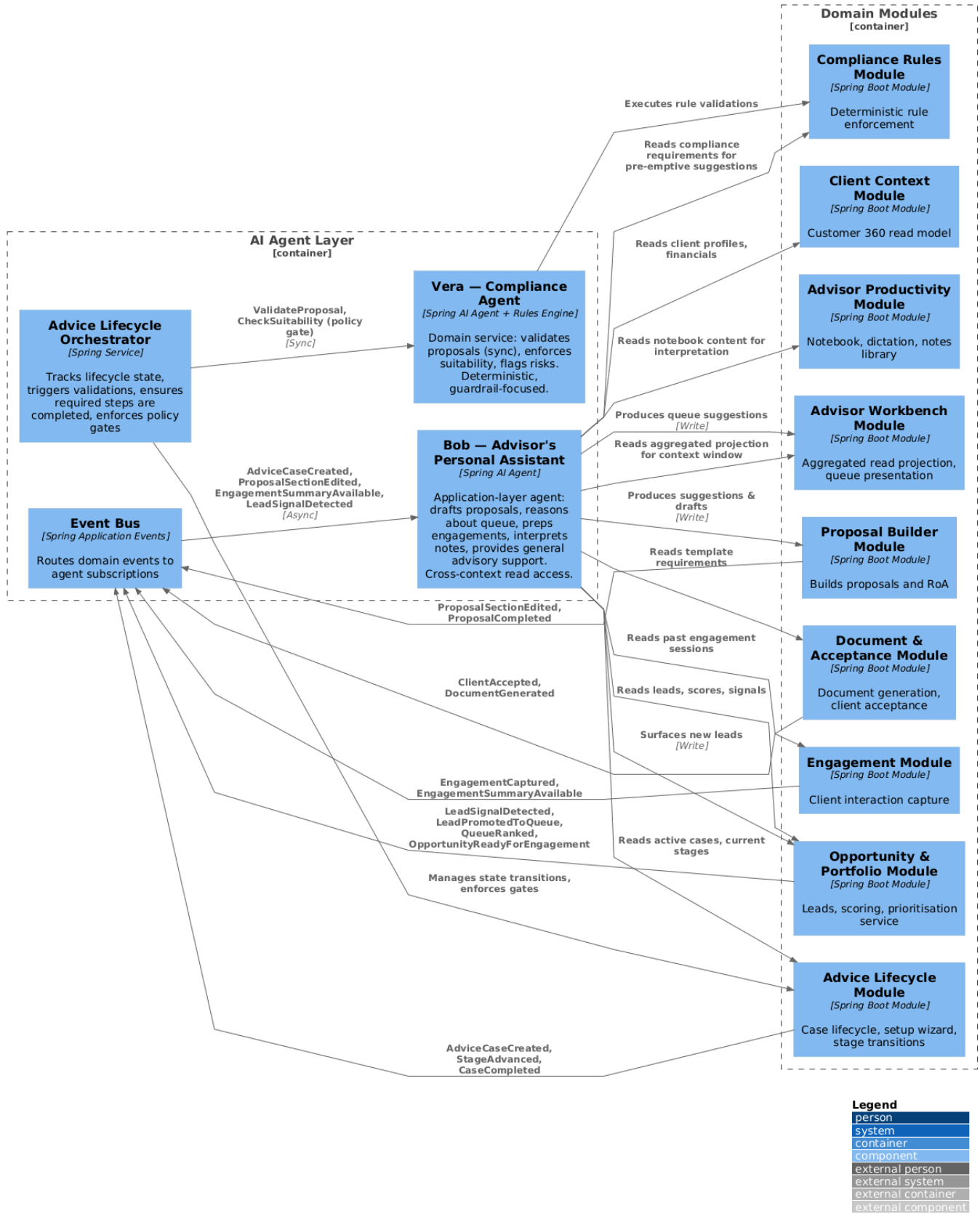
Shows the internal module structure of the monolith, aligned to the revised bounded contexts.



Level 3: Component Diagram – AI Agents & Orchestration

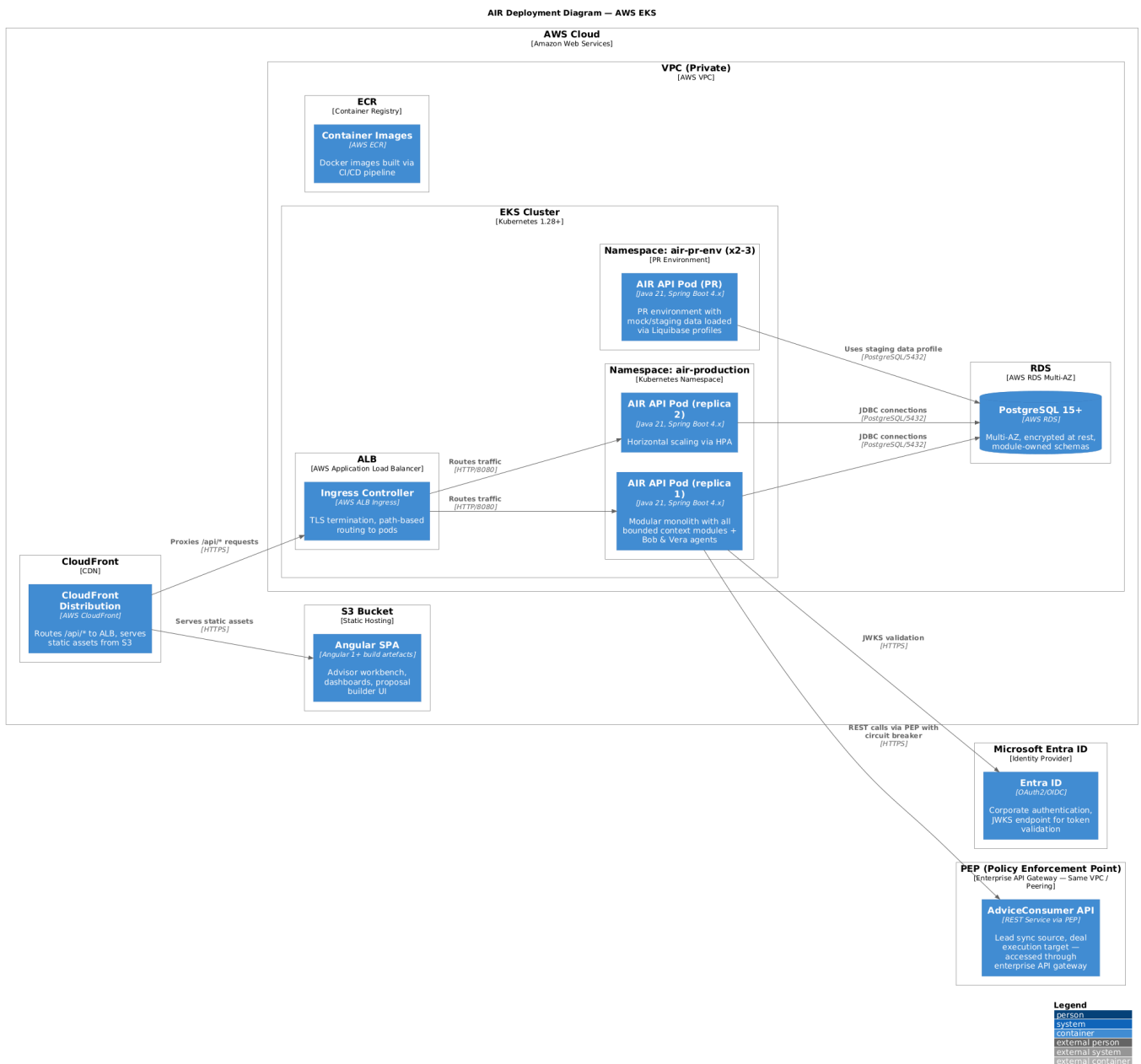
Focused view of how Bob and Vera interact with the domain modules. Gary is deferred.

AI Agents & Orchestration — Component Diagram



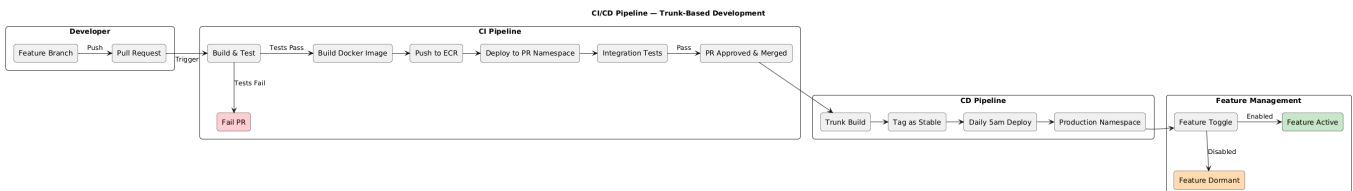
Level 4: Deployment Diagram

Shows the physical deployment topology on AWS.



CI/CD Pipeline Flow

Illustrates the trunk-based development and deployment pipeline.



Event Flow Diagram

Shows the event-driven orchestration pattern aligned to the revised bounded contexts.

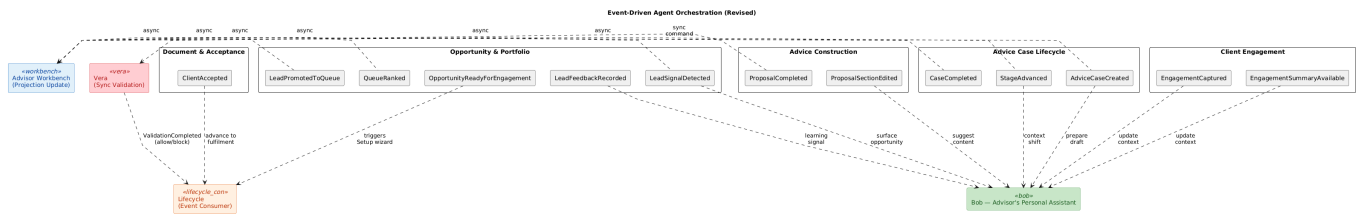


Diagram Summary

Level	Diagram	Purpose
C4 Level 1	System Context	Shows AIR in its ecosystem with users and external systems
C4 Level 2	Container	Shows deployable units – SPA, API monolith, DB, Bob & Vera agents, event bus
C4 Level 3	Component (API)	Shows internal bounded context modules within the monolith (9 active contexts)
C4 Level 3	Component (Agents)	Shows Bob's cross-context read/write access and Vera's sync validation role
C4 Level 4	Deployment	Shows AWS infrastructure topology – EKS, RDS, CloudFront, S3
Supplementary	CI/CD Pipeline	Shows trunk-based development and daily deployment flow
Supplementary	Event Flow	Shows event-driven orchestration with revised event catalogue

Key Architecture Characteristics

- **Modular Monolith:** All bounded contexts deployed as a single unit for speed, with explicit module boundaries for future decomposition
- **9 Active Bounded Contexts:** Advisor Workbench, Opportunity & Portfolio, Client Context, Advice Case Lifecycle, Advice Construction, Compliance Rules, Client Engagement, Document & Acceptance, Advisor Productivity
- **Bob as Application-Layer Agent:** Cross-context personal assistant with read access to all modules and write access to Proposal Builder, Workbench, and Opportunity
- **Vera as Domain Service:** Synchronous compliance validation within the Compliance Rules module; async monitoring deferred
- **Gary Deferred:** Professional coach agent deferred to future phase (Monitoring & Assurance context)

- **Prioritisation Service in Opportunity & Portfolio:** The Next-Best-Action ranking algorithm lives as a domain service within the Opportunity module, publishing ranked results to the Workbench projection
- **Opportunity Setup Wizard in Lifecycle:** Triggered by `OpportunityReadyForEngagement` event from Opportunity, owned by Advice Case Lifecycle
- **Event-Driven Backbone:** 13 domain events orchestrate cross-module communication and agent behaviour
- **Policy Gates:** Critical state transitions (e.g. `CompleteProposal`) go through synchronous Vera validation
- **AI Cannot Change State:** Agents suggest, flag, and advise – humans approve and transition
- **Feature Toggles:** Decouple code deployment from business release
- **PR Environments:** Ephemeral namespaces with mock data for rapid feedback
- **Module-Owned Data:** Each module manages its own schema via Liquibase

Alignment to Revised DDD

Bounded Context	Module	Classification	Notes
Advisor Workbench	<code>workbenchModule</code>	Core	New – owns queue presentation, scorecard, aggregated read projection
Opportunity & Portfolio	<code>opportunityModule</code>	Core	Owns leads, scoring, pipeline, prioritisation service, feedback
Client Context (Customer 360)	<code>clientModule</code>	Core (read model)	Read-only projection of upstream client systems
Advice Case Lifecycle	<code>lifecycleModule</code>	Core	Owns stages, transitions, setup wizard, outcome anchors
Advice Construction	<code>proposalModule</code>	Core	Owns proposals, RoA, versioning, diffs
Compliance Rules	<code>rulesModule</code>	Core	Sync validation only; async monitoring deferred
Client Engagement	<code>engagementModule</code>	Supporting	Structured interaction capture linked to cases
Document & Acceptance	<code>documentModule</code>	Supporting	Merged from Acceptance + Document Output
Advisor Productivity	<code>productivityModule</code>	Supporting	New – notebook, dictation, notes library

Deferred Contexts (not represented in current architecture)

Context	Reason	Future Home
Advisor Identity & Governance	Consumed from bank IAM (Entra ID)	External dependency
Compliance Monitoring	Vera's async mode not yet validated	Future module + event subscriptions
Monitoring & Assurance (Gary)	Coach agent not in MVP scope	Future module + agent
Deal Execution & Fulfilment	Absorbed into Lifecycle as terminal stage	Extract when complexity warrants

Glossary

Term	Definition
PEP	Policy Enforcement Point – the enterprise API gateway through which external service calls are routed
AIR	Advice & Intelligence Relationship – the platform being designed
Bob	AI personal assistant agent – advisor's cross-context co-pilot (proposal drafting, queue reasoning, engagement prep, notebook interpretation)
Vera	AI compliance agent – synchronous rule validation and suitability enforcement
Gary	AI professional coach agent – behavioural analysis and coaching insights (deferred)
RoA	Record of Advice – regulatory documentation of advice given
FNA	Financial Needs Analysis – client assessment inputs
HPA	Horizontal Pod Autoscaler – Kubernetes auto-scaling mechanism
NBA	Next-Best-Action – the prioritised recommendation for what an advisor should do next
QTD	Quarter-to-Date – performance measurement period
GAP	Pre-signature opportunity (not yet in pipeline)
Pipeline	Post-signature, pre-fulfilment opportunity (committed revenue)